

MIS 3116

Impact of Home Football Games

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Our Questions

How does Ames
already prepare for
Home games?

What causes the
difference in
crashes?

Does the number of
crashes increase on
home game days?



Datasets Used



Iowa Liquor Sales

[Iowa Liquor Sales](#)

This dataset is composed of all Sales to Class E liquor licensees. Class E licensees are grocery stores, convenience stores, etc. This means that this data is all wholesale/B2B transactions within the state of Iowa, which we filtered through to only include Ames.

Iowa Crash Data

[Iowa Crash Data](#)

This dataset includes crash data in the state of Iowa. It has crash severity ratings, the class or type of crash, as well as location of the crash. We filtered this data set by city to only include Ames.

Both datasets were filtered to include 2022-2025 to give us a good scope of recent history

Datasets Used

We made the CRASH_DATE and DATE column into datetime data type and made all numerical columns into float data type

```
df['CRASH_DATE'] = pd.to_datetime(df['CRASH_DATE'])  
display(df['CRASH_DATE'].head())
```

```
...          CRASH_DATE  
0  2023-01-26 00:00:00+00:00  
1  2023-01-27 00:00:00+00:00  
2  2023-01-26 00:00:00+00:00  
3  2023-01-27 00:00:00+00:00  
4  2023-01-28 00:00:00+00:00  
  
dtype: datetime64[ns, UTC]
```

```
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns  
df[numerical_cols] = df[numerical_cols].astype(float)  
display(df[numerical_cols].head())
```

Datasets Used

Game Day Calendar (2023–2025)

- **Date**
 - home game date
- **Opponent**
 - team played
- **Kickoff**
 - start time
- **is_cyhawk**
 - Cyhawk game
- **is_conference**
 - Big 12 opponent
- **game_type**
 - Early (<17:00) vs Night (≥17:00)

```
("2023-09-02", "Northern Iowa", "13:00", False, False),
("2023-09-09", "Iowa", "14:30", True, False), # Cy-Hawk!
("2023-09-23", "Oklahoma State", "~14:00", False, True),
("2023-10-07", "TCU", "19:00", False, True),
("2023-11-04", "Kansas", "~14:00", False, True),
("2023-11-18", "Texas", "19:00", False, True),

# --- 2024 ---
("2024-08-31", "North Dakota", "14:30", False, False),
("2024-09-14", "Arkansas State", "~14:00", False, False),
("2024-09-28", "Baylor", "18:36", False, True),
("2024-10-19", "UCF", "18:30", False, True),
("2024-11-02", "Texas Tech", "~14:00", False, True),
("2024-11-16", "Cincinnati", "~14:00", False, True),
("2024-11-30", "Kansas State", "18:30", False, True), # Farmageddon

# --- 2025 ---
("2025-08-30", "South Dakota", "~14:00", False, False),
("2025-09-06", "Iowa", "11:00", True, False), # Cy-Hawk!
("2025-09-27", "Arizona", "18:00", False, True),
("2025-10-25", "BYU", "14:30", False, True), # Homecoming
("2025-11-01", "Arizona State", "12:00", False, True),
("2025-11-22", "TCU", "~14:00", False, True),
], columns=["Date", "opponent", "kickoff", "is_cyhawk", "is_conference"])

games["Date"] = pd.to_datetime(games["Date"])
games["is_home"] = True
games["kickoff_clean"] = games["kickoff"].str.replace("~", "", regex=False)
games["kickoff_hr"] = pd.to_datetime(games["kickoff_clean"], format="%H:%M").dt.hour
games["kickoff_min"] = pd.to_datetime(games["kickoff_clean"], format="%H:%M").dt.minute
games["game_type"] = np.where(games["kickoff_hr"] < 17, "Early", "Night")
games["YearWeek"] = games["Date"].dt.to_period("W-SUN")
```

Our Hypothesis - Preparation

Business-to-business liquor sales increase on all home game weeks.

How We Scoped the Data

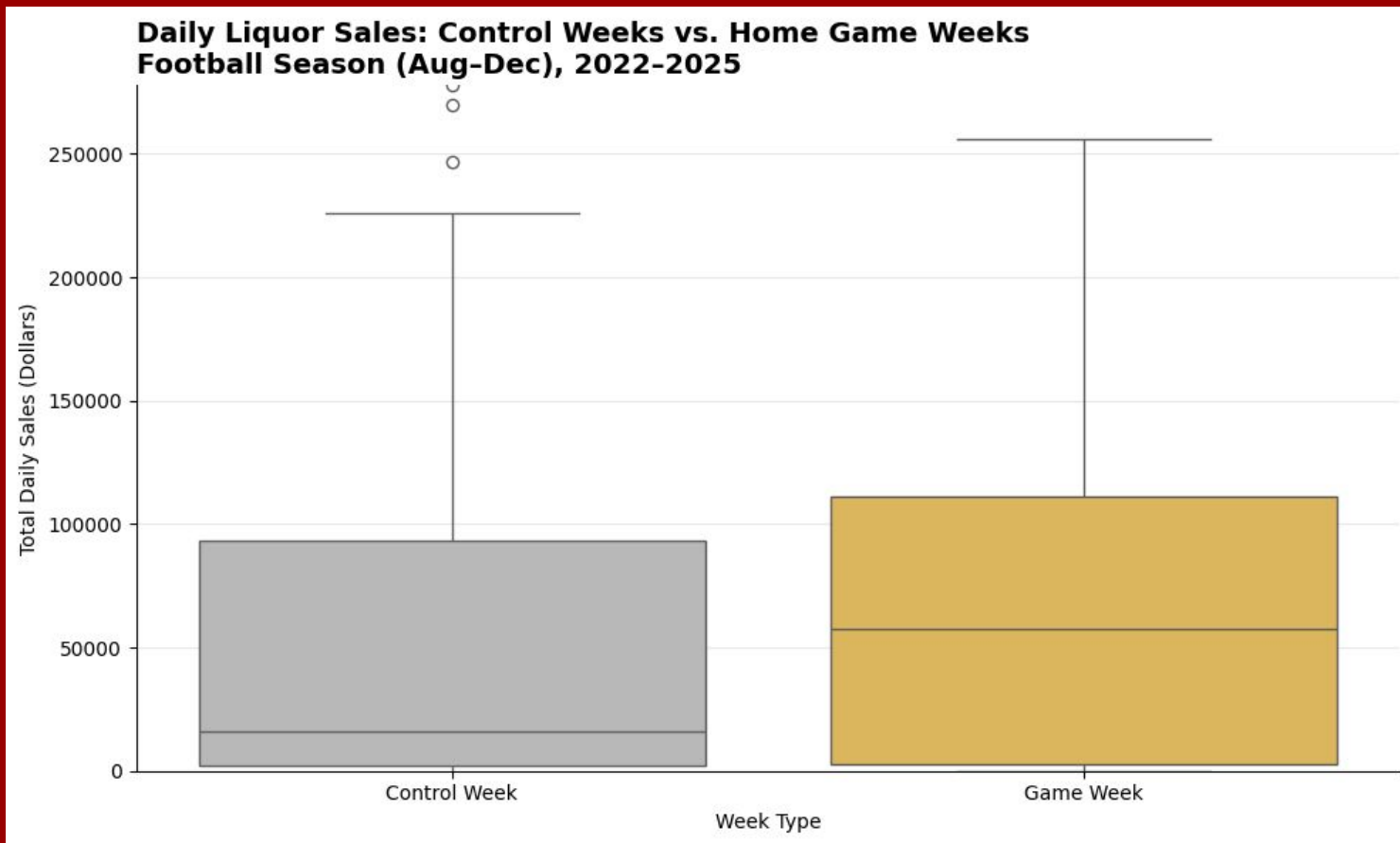
Sorting the Data Further

Narrowed to football season
(Aug-Dec)
Filters out holiday and seasonal
swings in alcohol sales, isolating
game-day effects.

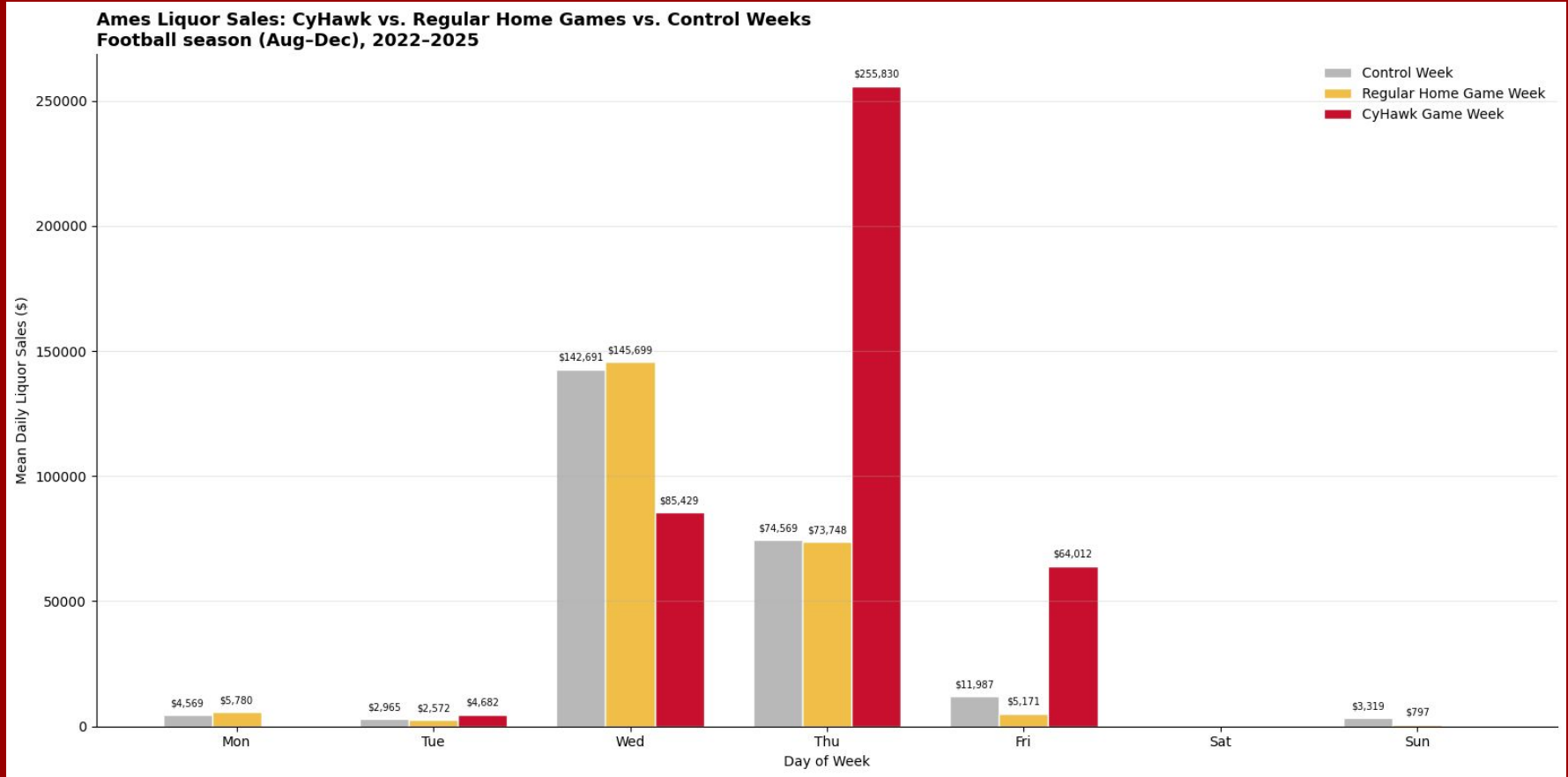
Collection/Lack of Data problem

Weekend data is sparse
Few transactions recorded on
Saturdays and Sundays - unclear
whether that's low purchasing or a
collection gap.

Preparation Findings



Preparation Findings



Preparation Findings

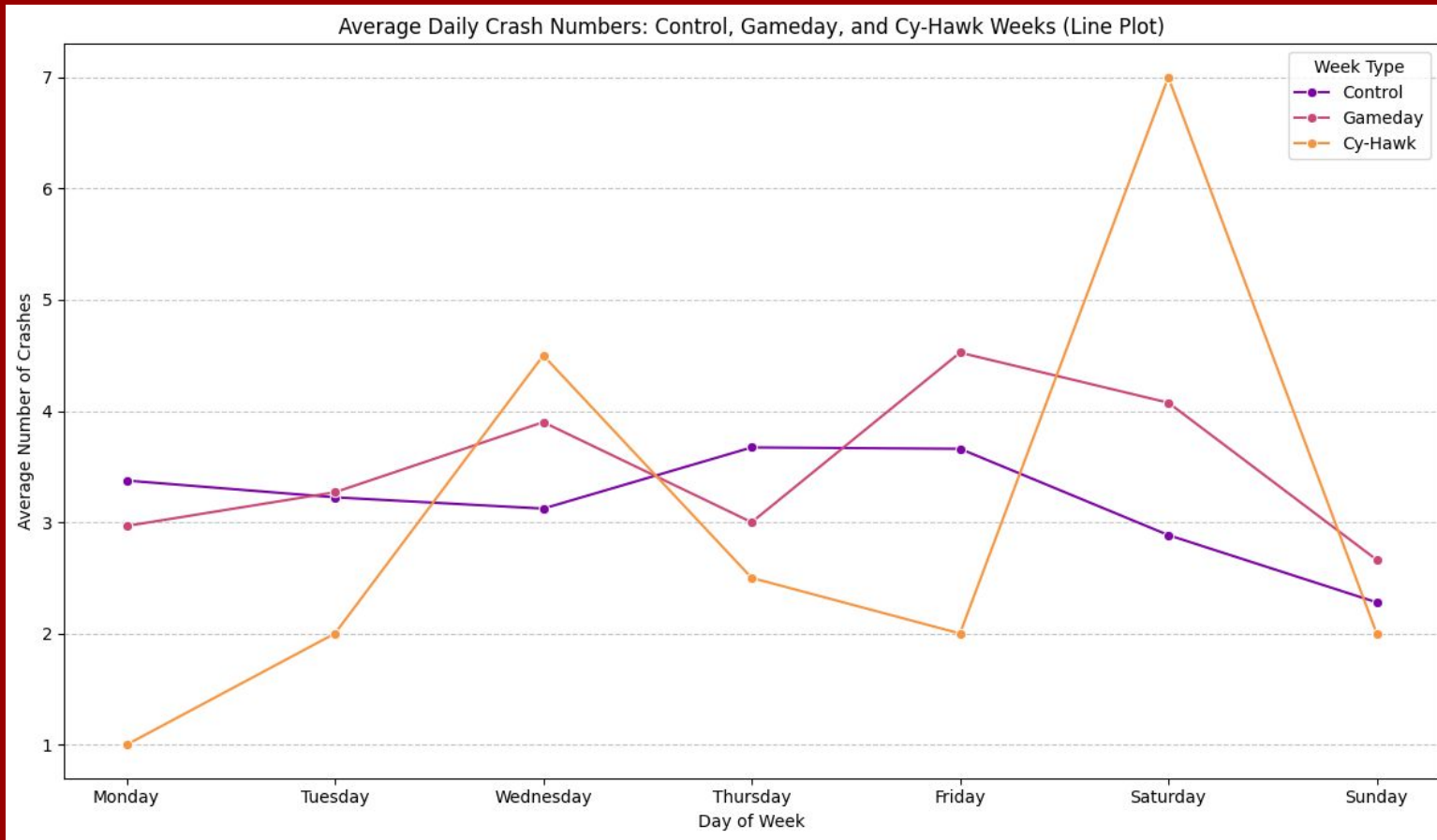
- On the average gameday, the total business-to-business liquor sales do not increase
- Stores heavily stock up for the Cy-Hawk game to provide for the larger crowd



Our Hypothesis - Crash Findings

The number of crashes will increase on Saturdays, especially because of drug or alcohol related incidents.

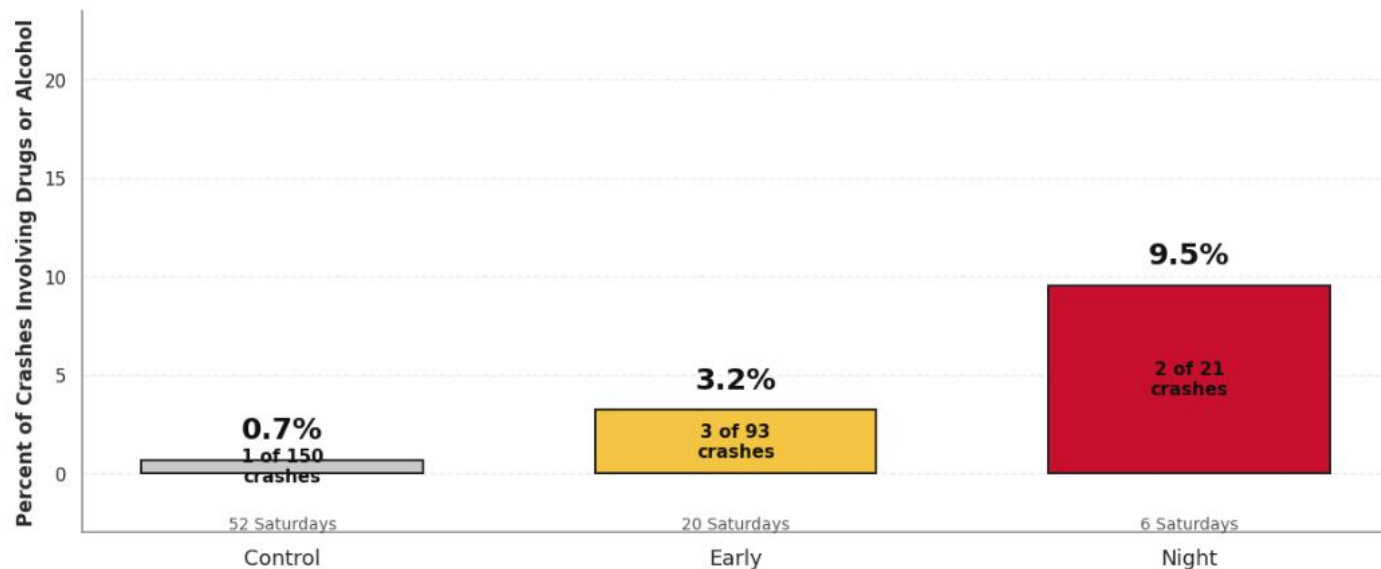
Crash Findings



Crash Findings

Night Home Games Show the Highest Drug/Alcohol-Related Crash Risk

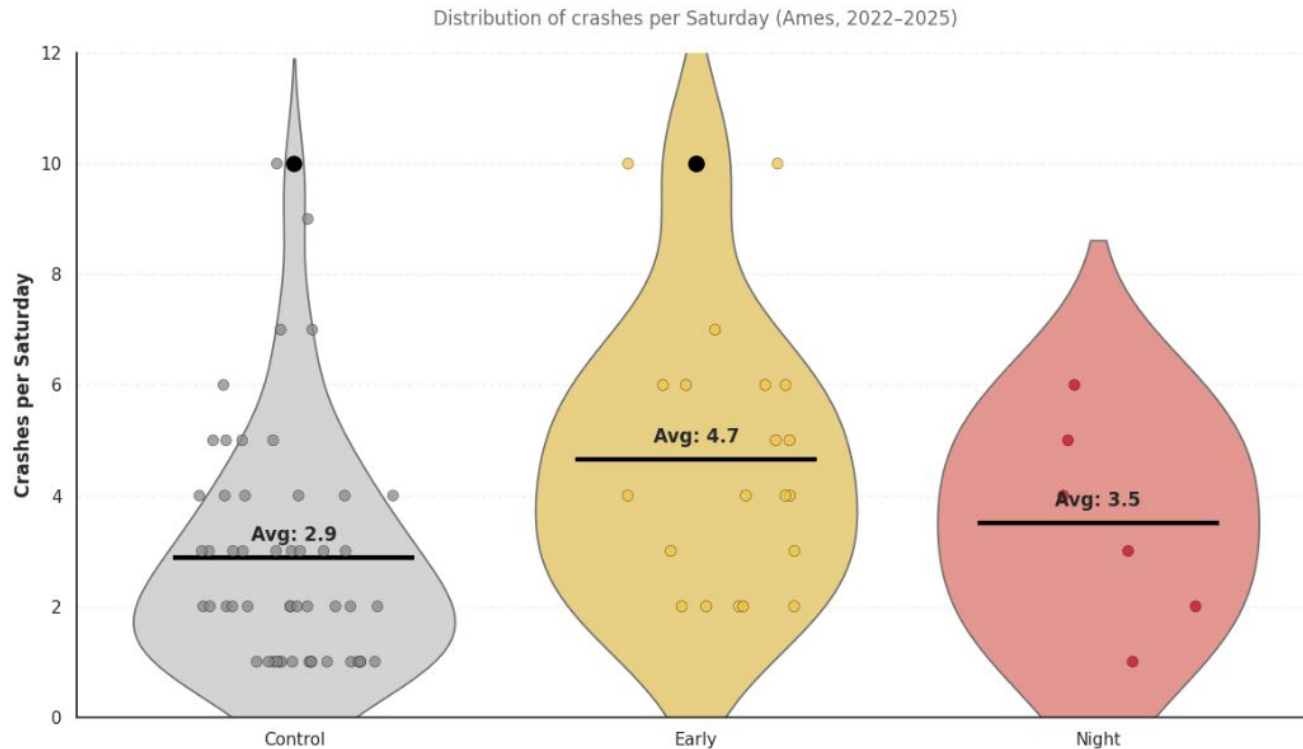
Ames Saturday crashes during ISU football season, 2022-2025



Note: Control = non-home-game Saturdays. Early and Night are based on ISU home football kickoff time.

Crash Findings

Night Games Create More Extreme and Unpredictable Crash Conditions



Crash Findings

- Night games are the primary cause of higher crash risk
- They increase alcohol-related crashes
- They lead to more severe and unpredictable outcomes
- Risk is driven by timing and behavior, not just traffic



Conclusion

Liquor Sales Only Spike for Cy-Hawk

Typical home games show very small B2B increase – CyHawk is the outlier.

More Crashes, Same Per-Capita Risk

Raw crash counts rise, but so does population. The rate stays comparable to control weekends.

Night Games = 13x More Impaired Drivers

9.5% of night-game Saturday crashes involve drugs/alcohol vs 0.7% on control days.

Recommendations

For ISU

- More DUI Patrols on Night Games
- Free Post-Game Shuttles
- Safety Alerts Tied to Kickoff Time

For Ames

- Normal Stock for Most Games
- Scale Traffic Resources by Game Type

